

Engineering Mathematics II

Learner Notes

Advanced construction calculations, calculus, vectors, matrices, and data interpretation.

Learning outcome	Focus
Apply advanced algebra and calculus	Applied mathematics in workplace and training contexts.
Use vectors and matrices	Applied mathematics in workplace and training contexts.
Analyze construction data	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Building Technology. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\frac{dy}{dx} = f'(x)$$

$$\int_a^b f(x) dx$$

$$\vec{R} = \vec{A} + \vec{B}$$

$$AX = B$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\frac{dy}{dx} = f'(x)$$

4. Practice Questions

1. Differentiate a cost function.

2. Calculate resultant forces in a simple frame.
3. Use matrices to solve building quantity equations.